# Supporting Information:

# Silent but deadly? Physics says seldom, memory says mostly

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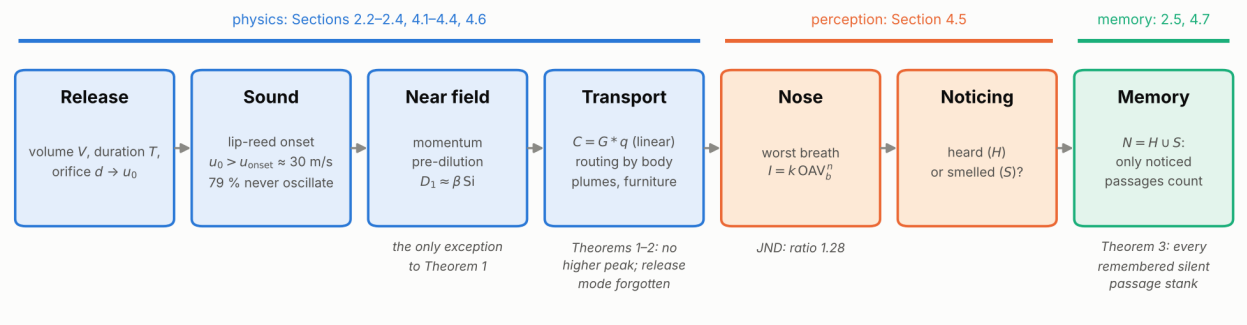



Figure S1: From release to memory: the quantity each stage passes on, and where Theorems 1–3 apply.

## S1 CFD validation

- *Free round jet* (axisymmetric, Re = 6700, three meshes). The velocity-decay constant is 5.0 and the scalar constant $K_c = 4.3$–$4.6$, with a grid uncertainty of at most 3.7%. The $k$–$\omega$ SST model spreads round jets about 25% too fast, which over-dilutes audible releases by about 15% and so favours the silent release.

- *Convective boundary layer* of a simplified seated manikin with skin at 33 °C, compared with thermal-manikin data.[S1] The face-front peak velocity is 40–56% too high at room temperatures of 20–26 °C; its change with room temperature is reproduced (−27% against −33% measured). The fast boundary layer shortens travel times to the releaser's own nose by about 30%, and the analytic model stretches the CFD travel times by a factor of 1.4 to correct for it.

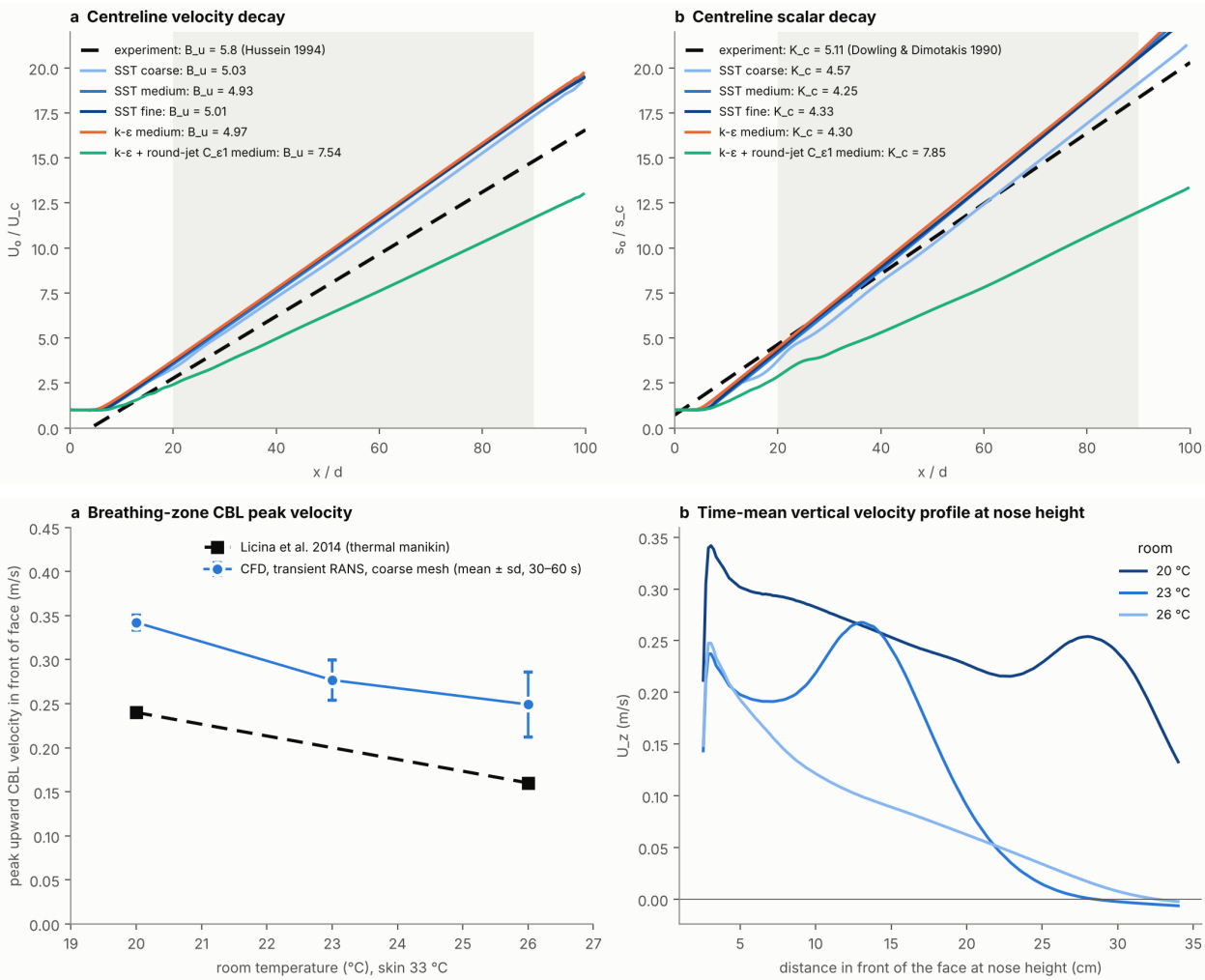



Figure S2: Validation. Top: free round jet at Re = 6700 on three meshes (centreline velocity and scalar decay, spreading). Bottom: face-front convective boundary layer of the seated manikin against thermal-manikin data[S1] at 20–26 °C, and time-mean vertical velocity profiles at nose height.

## S2 CFD scenes

All scenes use the solver of the main text, skin at 33 °C and walls at 23 °C. Each is first run without a release until the body plumes have developed (30 s; 90 s with ventilation). From that flow field the same 92 mL is released behind the releaser's buttocks, pointing down and back, as an inlet that matches the release's volume flow and momentum: silent, 2 s through 8 mm ($u_0 = 0.9\,\mathrm{m\,s^{-1}}$); audible, 0.5 s through 2.4 mm ($41\,\mathrm{m\,s^{-1}}$). The odorant is a passive scalar, sampled 2.5 cm in front of each nose, over 5 cm discs at nose height (breathing zone) and over sinusoidal breathing cycles; the worst breath is the most concentrated single inhalation.

- *Sofa.* A 3.2 × 3.2 × 2.6 m living room with a 0.5 m ceiling vent for pressure relief (size and position assumed). Two seated manikins sit 0.6 m apart on a two-seat sofa whose base is solid to the floor. Mesh 0.72 million cells, refined to 5 mm at the release and 2.5 mm on the inlet face. Runs of 30 s, the upright one extended to 120 s. Two variants rotate the releaser's upper body about the hip: reclined by 4.5 °, so that the back closes onto the backrest, and leaning forward by 10 °, so that the gap behind the back opens.

- *Elevator.* The car of a 630 kg, eight-person elevator, 1.10 m wide, 1.40 m deep and 2.10 m high (a manufacturer's standard car, within the maximum area of 1.66 $\mathrm{m^2}$ allowed for that load). Four standing manikins (nose at 1.59 m) stand in a 2 × 2 grid facing the door; the fan is off, and a 0.2 m ceiling opening relieves pressure (size assumed). Mesh 0.91 million cells. Runs of 60 s with the release behind a front-row or a back-row occupant; mirror symmetry gives the other two positions.

- *Standing pair.* The sofa room without the sofa, with two standing manikins face to face, noses 1.0 m apart and 0.8 m of open floor behind each back. Still air and office ventilation share one mesh of 0.43 million cells. Ventilation is one square ceiling

diffuser above the pair supplying $21.2\,\mathrm{L\,s^{-1}}$, i.e. 2.9 air changes per hour (EN 16798-1 Category II for two people in a low-polluting $10\,\mathrm{m^2}$ room: $7\,\mathrm{L\,s^{-1}}$ per person plus $0.7\,\mathrm{L\,s^{-1}\,m^{-2}}$[S2]). The air leaves the diffuser along the ceiling at an assumed $1.5\,\mathrm{m\,s^{-1}}$, at room temperature, and exits through a ceiling grille. Runs of 60 s.

# S3 Sofa

*Silent release.* The silent release stays in the convective boundary layer of the releaser's back. It rises in the gap between back and backrest, over the head and to the ceiling. Within 30 s its breathing-zone OAV is at most $10^{-10}$ at both noses, while within 25 cm of the releaser's head it reaches 6 000.

*Audible release.* The audible jet hits the corner between seat and backrest, spreads sideways along the gap and rises up both backs as a wider, more dilute sheet; part of it curls around the heads. It reaches the neighbour's head region within 6 s and both breathing zones at OAV 0.2–0.6.

*Reclined by 4.5 °.* With the back closed onto the backrest, the silent release leaves the seat-level wedge sideways and still rises behind the head; no breathing zone receives more than $10^{-10}$. The audible jet wraps around the releaser's face and exceeds threshold at their own nose (OAV 2–4).

*Leaning forward by 10 °.* With the gap behind the back open, both releases rise behind the head, and the releaser's nose, 13 cm further forward, receives neither (breathing-zone OAV $\leq 10^{-5}$). The silent cloud reaches the region within 25 cm of the neighbour's head as strongly as the audible one (OAV 580 against 450) but not the neighbour's breathing zone ($10^{-4}$); the audible jet reaches it at OAV 2.6.

*120 s.* Both clouds reach the ceiling, return through the room and arrive at both noses from about 45 s. After about 70 s the silent and audible traces at the neighbour are indistinguishable (worst breath 0.39 for both). At the releaser's own nose the mixed silent cloud

is 1.5–2 times higher and touches the trained-panel threshold for about a second near 100 s. These levels are 10–23 times below the thresholds of untrained observers.

Table S1: Sofa: peak breathing-zone OAV within 30 s of the release (mean flow).

| nose | posture | silent | audible |
|---|---|---|---|
| releaser | upright | $5 \times 10^{-11}$ | 0.46 |
| releaser | reclined 4.5 ° | $9 \times 10^{-11}$ | 3.3 |
| releaser | forward 10 ° | $3 \times 10^{-7}$ | $1.5 \times 10^{-5}$ |
| neighbour | upright | $3 \times 10^{-14}$ | 0.63 |
| neighbour | reclined 4.5 ° | $3 \times 10^{-15}$ | 0.19 |
| neighbour | forward 10 ° | $1 \times 10^{-4}$ | 2.6 |

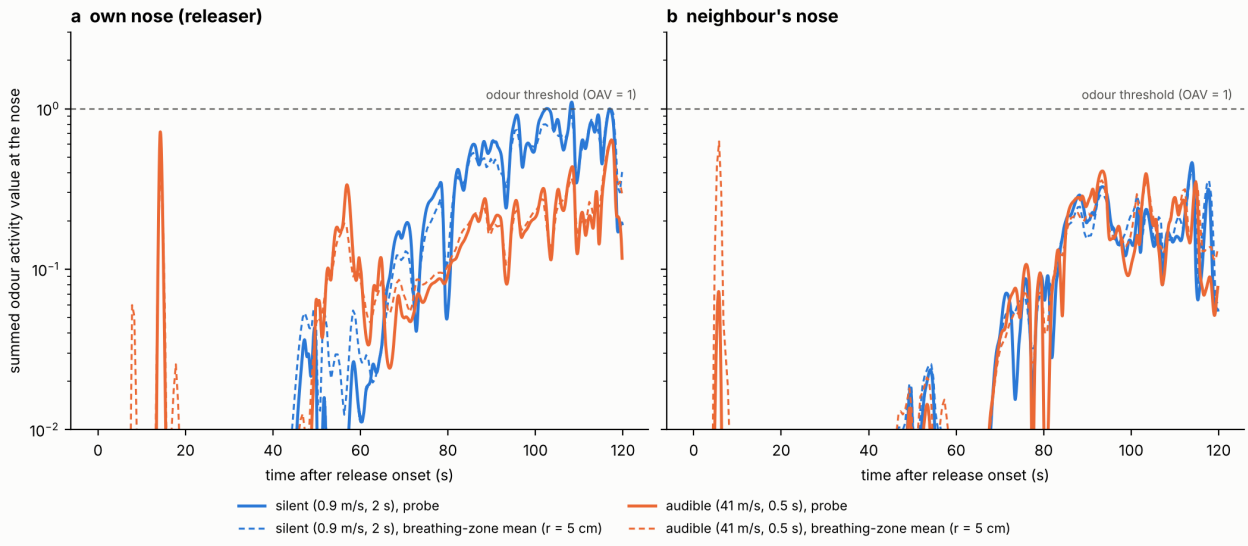



Figure S3: Sofa, upright: OAV at the releaser's and the neighbour's nose over 120 s for the same 92 mL released silently or audibly. Solid: 2.5 cm in front of the nose; dashed: mean over a 5 cm breathing-zone disc. The horizontal line is the trained-panel odour threshold.

# S4 Elevator

A silent release follows the releaser's own back plume, so what stands behind the releaser decides where it goes.

- With a person behind (front-row releaser), it rises in the 0.2 m gap between the releaser's back and that person's face, into their breathing zone. The silent release is

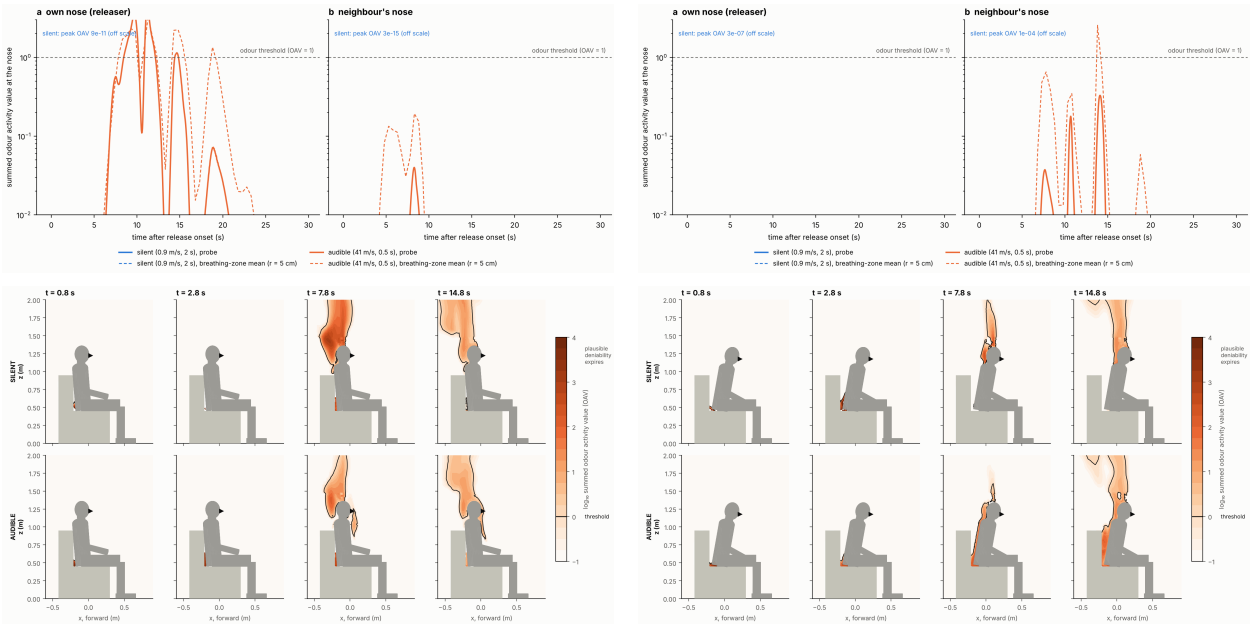



Figure S4: Sofa, releaser reclined by 4.5 ° (left) or leaning forward by 10 ° (right), 30 s. Top: OAV at both noses, as in Figure S3. Bottom: $\log_{10}$ OAV in the releaser's mid-sagittal plane, silent (upper rows) and audible (lower rows).

stronger at every nose ($R = 1.4$–$4.3$), most of all for the person behind (OAV 27).

- With the car wall behind (back-row releaser), it rises between back and wall and reaches the front row only below threshold. The audible jet hits the floor, is lifted by all four body plumes, and is stronger at every nose ($R = 0.20$–$0.55$).

With a front-row releaser the person behind reaches the trained-panel threshold first (8.8 s), before the releaser (17.6 s); with a back-row releaser the releaser is first. Whoever smells it first is therefore the releaser in half of the positions, for both modes.

*Identifying the releaser.* Each occupant other than the releaser reacts with the detection probability of the main text's observer models, independently. Onlookers see only who reacted and infer the releaser with a uniform prior, for a releaser who never reacts and one who reacts like the others. With distracted occupants no one reacts in 87–92% of passages, and the releaser is identified at 0.28–0.31; attentive occupants identify the releaser at 0.39–0.52 (chance 0.25). Silent and audible passages are identified about equally well by odour; the sound of an audible passage, which also points to the releaser, is not modelled.

Table S2: Elevator: expected worst-breath OAV within 60 s at each nose (trained-panel thresholds; mean flow), for a front-row and a back-row releaser.

| releaser | nose | silent | audible | $R$ |
|---|---|---|---|---|
| front row | releaser | 6.7 | 1.9 | 3.4 |
| | beside | 3.6 | 0.8 | 4.3 |
| | behind | 27 | 20 | 1.4 |
| | diagonal | 7.6 | 2.7 | 2.9 |
| back row | releaser | 4.5 | 23 | 0.20 |
| | beside | 2.3 | 5.9 | 0.39 |
| | in front | 0.57 | 2.8 | 0.20 |
| | diagonal | 0.41 | 0.74 | 0.55 |

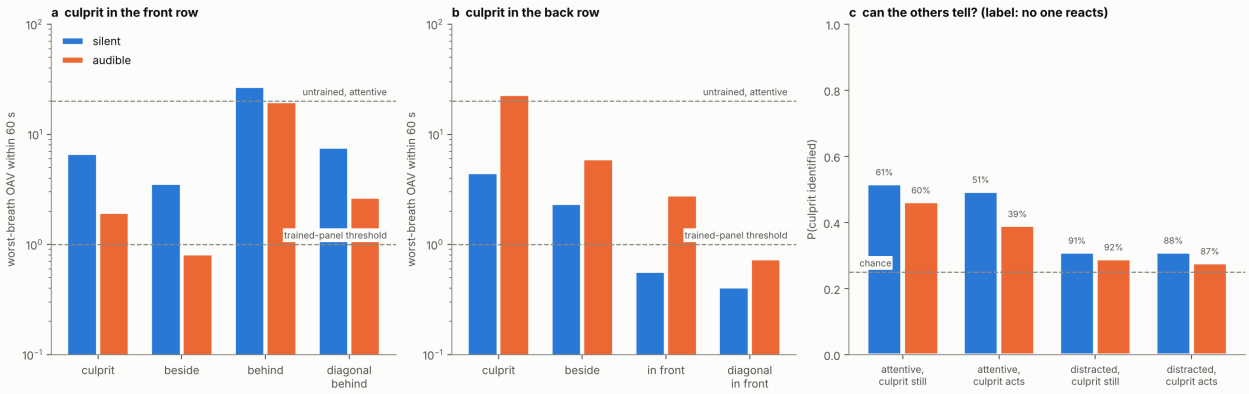



Figure S5: Elevator: four standing occupants in a closed 630 kg car, fan off, same 92 mL. (a, b) Expected worst-breath OAV within 60 s at each nose for a front-row and a back-row releaser; dashed lines mark the trained-panel threshold and the attentive population threshold. (c) Probability that the others identify the releaser from who reacted, for attentive and distracted onlookers and a releaser who stays still or reacts along; labels give the probability that no one reacts. The dashed line is chance.

# S5 Standing pair

Results pending (simulations running).

# S6 Breath sampling

An audible puff reaches the neighbour after a median of 16 s and passes with a time scale of about 3.9 s (10–90%: 1.5–12 s). Without a warning, the breathing phase barely matters: for a median passage the worst breath captures 81–99% of the peak. A listener warned by the sound needs about 25 s of breath-hold for an even chance of missing the puff (worst breath below 10% of the peak); 33 s, the typical maximum at end-expiration, gives 62%.

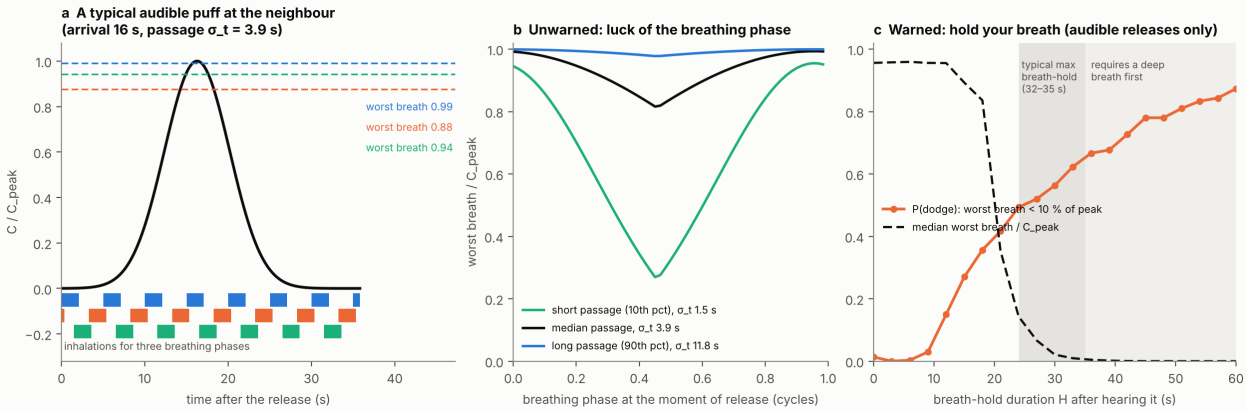



Figure S6: Breath sampling at the neighbour's nose for audible releases. (a) A typical puff (arrival 16 s, passage time scale 3.9 s) and the inhalations of three breathing phases, with the worst breath of each. (b) Without warning: worst breath relative to the peak against the breathing phase at release, for short, median and long passages. (c) Warned: probability of missing the puff (worst breath below 10% of the peak) and median worst breath, against the breath-hold duration after hearing the release (0.5 s reaction latency).

# S7 Cross-cultural survey

We searched dictionaries, proverb collections and etymological handbooks in 32 languages or historical stages for a dedicated word for a silent passage and for an explicit claim that it smells worse, and checked every entry used against a retrieved source. Each language

is graded by its strongest evidence: A, an explicit saying (traditional, or in media or film only); B, a dedicated word defined by its smell; C, a dedicated word without a smell claim; D, nothing found. The silent/loud distinction is lexicalised in at least five language families. The explicit claim is traditional only in Chinese and English, and smell-defined silent words arose independently at least four times within Indo-European. The survey records whether the belief is documented in a language, not how many speakers hold it.

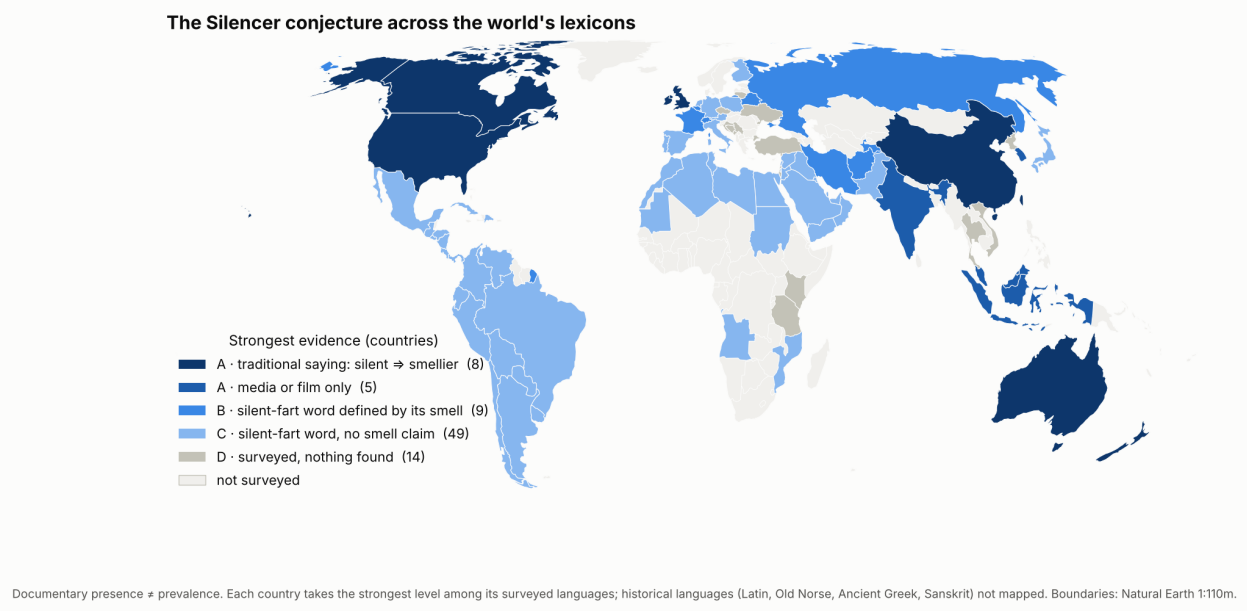



Figure S7: The belief across languages, by evidence level. Each country takes the strongest level among its surveyed languages. A: an explicit saying that silent passages smell worse (traditional, or in media or film only); B: a dedicated word for a silent passage defined by its smell; C: a dedicated word with no smell claim; D: surveyed, nothing found.

# S8 Code and data

The archive contains the code, inputs and outputs of every result; Table S3 lists where each comes from.

# S9 Parameters

Table S3: Source files of the results in the archive.

| result | files |
|---|---|
| parameters (Table S4) | `params/params.yaml` |
| odorant shares, buoyancy, losses | `chem/odorants.py`, `chem/buoyancy.py`, `chem/reactivity.py`, `chem/chem_report.md` |
| acoustics | `acoustics/regime.py`, `acoustics/run_m2.py`, `acoustics/m2_report.md` |
| analytic dispersion model | `gate0/model.py`, `gate0/run_gate0.py`, `gate0/gate0_report.md` |
| uncertainty, regime map | `uq/run_m7.py`, `uq/m7_report.md` |
| passages as emitted in the scenes | `uq/routing_headline.py`, `uq/routing_headline.json`, `uq/fig_routing.py` |
| CFD validation | `cfd/validation/jet`, `cfd/validation/cbl`, `cfd/validation/m3_report.md` |
| sofa | `cfd/cases/sofa`, `cfd/cases/sofa/m4_report.md` |
| elevator | `cfd/cases/elevator`, `cfd/cases/elevator/m5_report.md` |
| standing pair | `cfd/cases/standing` |
| geometry | `cfd/geometry/manikin.py` |
| breath sampling, perception | `perception/model.py`, `perception/run_m6.py`, `perception/m6_report.md` |
| observer models, memory | `uq/observer_ps.py`, `perception/selection.py`, `perception/run_selection.py` |
| cross-cultural survey | `culture/cross_cultural_table.csv`, `culture/survey_summary.md` |

Table S4: Model parameters: nominal value, range sampled or tested, and basis: taken from a cited measurement, derived from cited data, partly sourced, taken from a non-peer-reviewed source, or assumed. Parameter names are those of the code; the source and notes for every entry are in `params/params.yaml` of the code archive.

| parameter | nominal | range | units | basis |
|---|---|---|---|---|
| *Source* | | | | |
| `T_gas` | 310.15 | 309.85–310.65 | K | measured |
| `V_passage` | $9.2 \times 10^{-5}$ | $1.7 \times 10^{-5}$–$3.75 \times 10^{-4}$ | $m^3$ | measured |
| `V_daily` | $7.05 \times 10^{-4}$ | $4.76 \times 10^{-4}$–0.001491 | $m^3\,d^{-1}$ | measured |
| `n_daily` | 10 | 1–20 | $d^{-1}$ | measured |
| `d_orifice` | 0.004 | 0.001–0.008 | m | partly sourced |
| `T_release` | 1 | 0.3–20 | s | partly sourced |
| `dp_rectal` | 2000 | 500–4000 | Pa | assumed |
| `p_anal_rest` | 8666 | 6133–12000 | Pa | measured |
| `g_reduced` | 1.7 | -0.66–4.68 | $m\,s^{-2}$ | derived |
| `nu_gas` | $1.7 \times 10^{-5}$ | | $m^2\,s^{-1}$ | assumed |
| `c_H2S` | 1.06 | 0.26–1.86 | $\mu mol\,L^{-1}$ | measured |
| `c_CH3SH` | 0.21 | 0.05–0.37 | $\mu mol\,L^{-1}$ | measured |
| `c_DMS` | 0.08 | 0.04–0.12 | $\mu mol\,L^{-1}$ | measured |
| `sulfur_sigma_ln` | 1.1 | 1–1.25 | – | derived |
| *Odour thresholds* | | | | |
| `H2S` | 0.41 | 0.41–0.9 | ppbv | measured |
| `CH3SH` | 0.07 | | ppbv | measured |
| `DMS` | 3 | | ppbv | measured |
| *Jet and puff* | | | | |
| `B_u` | 5.8 | 5.8–5.9 | – | measured |
| `S_jet` | 0.094 | 0.094–0.102 | – | measured |
| `K_c` | 5.11 | 4.73–5.2 | – | measured |
| `Gamma_sj` | 2.8 | 2.74–2.86 | – | measured |

Table S4: Model parameters (continued).

| parameter | nominal | range | units | basis |
|---|---|---|---|---|
| C_puff | 2.6 | 2.54–2.66 | – | measured |
| alpha_th | 0.25 | 0.15–0.35 | – | measured |
| alpha_jet | 0.07 | 0.065–0.08 | – | measured |
| alpha_plume | 0.12 | 0.1–0.16 | – | measured |
| *Environment* | | | | |
| u_handoff | 0.15 | 0.05–0.25 | $\mathrm{m\,s^{-1}}$ | assumed |
| u_CBL | 0.17 | 0.11–0.24 | $\mathrm{m\,s^{-1}}$ | measured |
| u_CBL_pelvis | 0.1 | 0.05–0.15 | $\mathrm{m\,s^{-1}}$ | assumed |
| u_own_eff | 0.11 | 0.07–0.16 | $\mathrm{m\,s^{-1}}$ | derived |
| delta_CBL | 0.1 | 0.05–0.15 | m | assumed |
| u_room | 0.08 | 0.03–0.15 | $\mathrm{m\,s^{-1}}$ | assumed |
| K_turb | 0.002 | $1 \times 10^{-4}$–0.013 | $\mathrm{m^2\,s^{-1}}$ | measured |
| D_H2S_air | $1.65 \times 10^{-5}$ | $1.62 \times 10^{-5}$–$1.77 \times 10^{-5}$ | $\mathrm{m^2\,s^{-1}}$ | partly sourced |
| *Geometry* | | | | |
| eye_sit_m | 0.804 | 0.75–0.86 | m | measured |
| eye_sit_f | 0.748 | 0.698–0.799 | m | measured |
| eye_stand_m | 1.64 | 1.536–1.753 | m | derived |
| eye_stand_f | 1.517 | 1.421–1.626 | m | derived |
| L_own | 0.74 | 0.65–0.85 | m | assumed |
| L_neighbour | 1 | 0.8–1.4 | m | assumed |
| *Breathing* | | | | |
| V_tidal | $6.3 \times 10^{-4}$ | $4 \times 10^{-4}$–$8.3 \times 10^{-4}$ | $\mathrm{m^3}$ | measured |
| f_breath | 12 | 10–16 | $\mathrm{min^{-1}}$ | measured |
| Ti_Ttot | 0.42 | 0.34–0.5 | – | measured |
| T_insp | 2.1 | 1.6–2.7 | s | derived |
| *Acoustics* | | | | |

Table S4: Model parameters (continued).

| parameter | nominal | range | units | basis |
|---|---|---|---|---|
| f0_flatus | 160 | 60–300 | Hz | partly sourced |
| bg_library | 40 | 35–45 | dB(A) | assumed |
| bg_living | 35 | 30–45 | dB(A) | assumed |
| bg_office | 45 | 43–55 | dB(A) | measured |
| bg_restaurant | 74 | 67–83 | dB(A) | measured |
| bg_subway | 77 | 74–80 | dB(A) | measured |
| P_th | 1200 | 600–2500 | Pa | partly sourced |
| C_d | 0.85 | 0.6–1 | – | partly sourced |
| alpha_mod | 0.7 | 0.5–1 | – | assumed |
| harm_slope | -12 | -18—6 | dB/octave | assumed |
| n_harm | 8 | | – | assumed |
| A_cloth | 1 | 0–3 | dB | partly sourced |
| r_listener | 0.6 | 0.4–3 | m | assumed |
| *Perception and observers* | | | | |
| n_stevens | 0.28 | 0.2–0.45 | – | measured |
| k_w_H2S | 0.95 | 0.8–1.2 | intensity units per decade | partly sourced |
| weber_C | 0.28 | 0.05–0.4 | $\Delta C/C$ | measured |
| tau_adapt | 80 | 40–120 | s | assumed |
| a_inf | 0.35 | 0.25–0.4 | – | partly sourced |
| sniff_ratio | 0.64 | 0.61–0.75 | – | measured |
| t_react | 0.5 | 0.3–1.5 | s | assumed |
| H_max | 33 | 24–90 | s | measured |
| thr_scale_typ | 0.05 | 0.05–0.1 | – | measured |
| thr_sd_bs | 2 | | binary steps | measured |
| distracted_factor | 26 | | – | measured |
| distracted_sd_bs | 3.5 | | binary steps | measured |

Table S4: Model parameters (continued).

| parameter | nominal | range | units | basis |
|---|---|---|---|---|
| `recovery_gain` | 1.5 | 1–2 | – | assumed |

# References


(S1) Licina, D.; Pantelic, J.; Melikov, A.; Sekhar, C.; Tham, K. W. Experimental investigation of the human convective boundary layer in a quiescent indoor environment. *Building and Environment* **2014**, *75*, 79–91.

(S2) Olesen, B. W. European standardisation on indoor air quality. *REHVA Journal* **2024**, 12–16.